King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 102 Major Exam I Term 231

Original exam date: 2 October 2023. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. If $\int_{-3}^{3} f(x)\,dx + \int_{3}^{6} f(x)\,dx - \int_{a}^{b} f(x)\,dx = \int_{-1}^{6} f(x)\,dx$, then $a - b =$

(a) -4

(b) -6

(c) 2

(d) -2

(e) 4

2. $\int_{0}^{5} |2x - 5|\,dx =$

(a) $\dfrac{23}{2}$

(b) $\dfrac{21}{2}$

(c) $\dfrac{27}{2}$

(d) $\dfrac{25}{2}$

(e) $\dfrac{29}{2}$

3. $\int_{0}^{\ln 2} 2e^{-x} \cosh x\,dx =$

(a) $\dfrac{3}{8} + \ln 3$

(b) $\dfrac{3}{8} - \ln 4$

(c) $\dfrac{3}{8} + \ln 2$

(d) $\dfrac{1}{4} + \ln 3$

(e) $\dfrac{1}{4} - \ln 2$

4. The Riemann sum for $f(x) = x^2+3x$ over the interval $[0, 8]$ where $x_0 = 0,\ x_1 = 1,\ x_2 = 3,\ x_3 = 7$ and $x_4 = 8$ and where $c_1 = 1,\ c_2 = 2,\ c_3 = 5$, and $c_4 = 8$, is equal to
(Note: c_i is a point in the ith subinterval)

(a) 273

(b) 270

(c) 271

(d) 272

(e) 274

5. The average value of the function $f(x) = \sec\left(\frac{\pi x}{6}\right)$ over $[0, 2]$ is equal to

(a) $\frac{2}{\pi}\ln(2+\sqrt{3})$

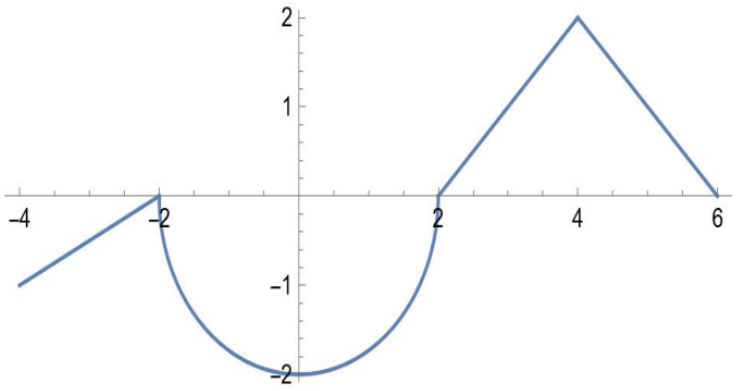
(b) $\frac{2}{\pi}\ln(2-\sqrt{3})$

(c) $\frac{3}{\pi}\ln(3-\sqrt{2})$

(d) $\frac{3}{\pi}\ln(2+\sqrt{3})$

(e) $\frac{3}{\pi}\ln(3+\sqrt{2})$

6. Using the attached graph, $\int_{-4}^{6} f(x)\,dx =$



(a) $2-2\pi$

(b) $3-2\pi$

(c) $3+2\pi$

(d) $4+2\pi$

(e) $4-2\pi$

7. $\int \sqrt{e^t - 3}\, dt =$

(a) $2\sqrt{e^t - 3} + \sqrt{3}\tan^{-1}\sqrt{\dfrac{e^t - 3}{3}} + c$

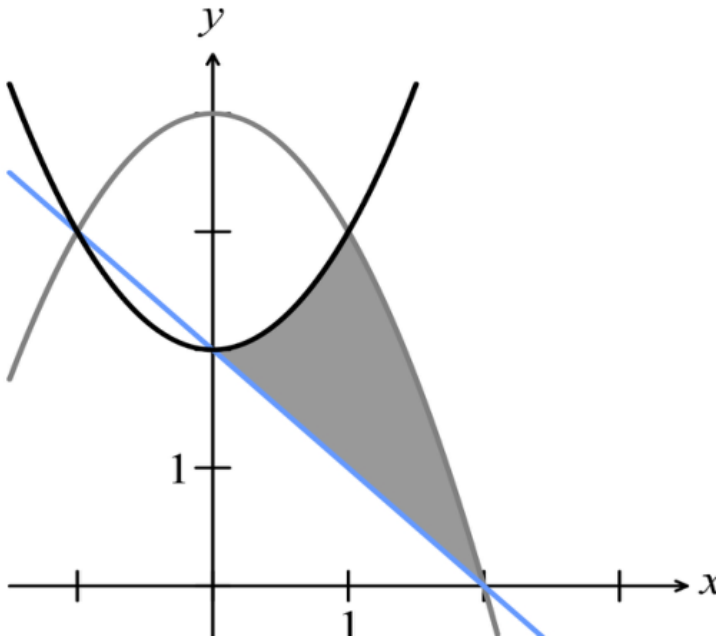
(b) $2\sqrt{e^t - 3} - 2\sqrt{3}\tan^{-1}\sqrt{\dfrac{e^t + 3}{3}} + c$

(c) $\tan^{-1}\sqrt{\dfrac{e^t - 3}{3}} + c$

(d) $2\sqrt{e^t - 3} - 2\sqrt{3}\tan^{-1}\sqrt{\dfrac{e^t - 3}{3}} + c$

(e) $\sqrt{e^t - 3} - \sqrt{3}\tan^{-1}\sqrt{\dfrac{e^t - 3}{3}} + c$

8. The area of the shaded region bounded by the graphs of $y_1 = x^2 + 2$, $y_2 = 4 - x^2$ and $y_3 = 2 - x$, as shown in the figure, is



(a) 1

(b) 2

(c) 3

(d) 5

(e) 4

9. If $F(x) = \int_0^{x^3} \sin t^2\, dt$, then $F'(1) =$

(a) 0

(b) $3 \sin 1$

(c) $\sin 1$

(d) $2 \sin 1$

(e) $4 \sin 1$

10. $\int (x+1)\sqrt{2-x}\, dx =$

(a) $\frac{2}{5}(2-x)^{3/2}(x-1) + c$

(b) $\frac{-2}{5}(2-x)^{1/2}(x+3) + c$

(c) $\frac{2}{5}(2-x)^{1/2}(x-3) + c$

(d) $\frac{-2}{5}(2-x)^{3/2}(x+3) + c$

(e) $\frac{-2}{5}(2-x)^{3/2}(x-3) + c$

11. $\int_1^9 \frac{dx}{\sqrt{x}\,(1+\sqrt{x})^2} =$

(a) $\frac{-1}{2}$

(b) 0

(c) 1

(d) $\frac{3}{2}$

(e) $\frac{1}{2}$

12. Let the velocity function $v(t) = t^2 - t - 12$, in feet per second, be given for a particle moving along a straight line, where t is the time in seconds. The total distance (in feet) that the particle travels over the interval $[1, 5]$ is equal to

(a) $\frac{65}{3}$

(b) $\frac{77}{3}$

(c) $\frac{79}{3}$

(d) $\frac{80}{3}$

(e) $\frac{56}{3}$

13. $\int_0^2 \dfrac{x^2-2}{x+1}\,dx =$

(a) $\ln 2$

(b) $\ln 3$

(c) $-\ln 2$

(d) $-\ln 3$

(e) 1

14. If we use the Midpoint Rule with $n = 4$ to approximate the area of the region bounded by the graph of the function $f(x) = x^2 + 4x$ and the x-axis over the interval $[0, 4]$, then we get

(a) 54

(b) 56

(c) 52

(d) 55

(e) 53

15. Using the summation formulas for $n = 10$, $\sum_{k=1}^{n} \dfrac{6k(k-1)}{n^3} =$

(a) 1.90

(b) 1.98

(c) 1.96

(d) 1.97

(e) 1.99

16. The area of the region bounded by the graphs of $f(y) = y(2-y)$ and $g(y) = -y$ is

(a) $\dfrac{13}{2}$

(b) $\dfrac{9}{2}$

(c) $\dfrac{7}{2}$

(d) $\dfrac{11}{2}$

(e) $\dfrac{5}{2}$

17. $\int e^{-x} \tan\left(e^{-x}\right) dx =$

(a) $\ln|\cos(e^{-x})| + c$

(b) $\ln|\tan(e^{-x})| + c$

(c) $\ln|\sin(e^{x})| + c$

(d) $\ln|\cos(e^{x})| + c$

(e) $\ln|\sin(e^{-x})| + c$

18. $\int_2^3 \dfrac{2x-3}{\sqrt{4x-x^2}}\, dx =$

(a) $1 - \sqrt{3} + \dfrac{\pi}{6}$

(b) $2 - 4\sqrt{3} + \dfrac{\pi}{6}$

(c) $4 + 2\sqrt{3} - \dfrac{\pi}{6}$

(d) $4 - 2\sqrt{3} + \dfrac{\pi}{3}$

(e) $4 - 2\sqrt{3} + \dfrac{\pi}{6}$

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	Combine: $\int_a^b f = \int_{-3}^{-1} f$, so $a=-3,\ b=-1,\ a-b=-2$.
2	**(d)**	Split at $x=\frac{5}{2}$: $\frac{25}{4}+\frac{25}{4}=\frac{25}{2}$.
3	**(c)**	$2e^{-x}\cosh x = 1+e^{-2x}$; integral $= \ln 2 + \frac{3}{8}$.
4	**(d)**	$4\cdot 1 + 10\cdot 2 + 40\cdot 4 + 88\cdot 1 = 272$.
5	**(d)**	$\frac{1}{2}\cdot\frac{6}{\pi}\ln\lvert\sec u+\tan u\rvert\Big\rvert_0^{\pi/3} = \frac{3}{\pi}\ln(2+\sqrt{3})$.
6	**(b)**	Signed areas: -1 (triangle) -2π (half disk) $+4$ (triangle) $= 3-2\pi$.
7	**(d)**	$u=\sqrt{e^t-3}$: $\int \frac{2u^2}{u^2+3}du = 2u - 2\sqrt{3}\tan^{-1}\frac{u}{\sqrt{3}}$.
8	**(b)**	Split at $x=1$: $\int_0^1(x^2+x)dx + \int_1^2(2+x-x^2)dx = \frac{5}{6}+\frac{7}{6}=2$.
9	**(b)**	$F'(x)=3x^2\sin x^6$, so $F'(1)=3\sin 1$.
10	**(d)**	$u=2-x$: $\frac{2}{5}u^{5/2}-2u^{3/2} = -\frac{2}{5}(2-x)^{3/2}(x+3)$.
11	**(e)**	$u=1+\sqrt{x}$: $\int_2^4 2u^{-2}du = \left[-\frac{2}{u}\right]_2^4 = \frac{1}{2}$.
12	**(c)**	$v<0$ on $[1,4]$, $v>0$ on $[4,5]$: $\frac{135}{6}+\frac{23}{6}=\frac{79}{3}$.
13	**(d)**	$\frac{x^2-2}{x+1} = x-1-\frac{1}{x+1}$; result $-\ln 3$.
14	**(e)**	$\Delta x=1$, midpoints $0.5, 1.5, 2.5, 3.5$: $2.25+8.25+16.25+26.25=53$.
15	**(b)**	$\frac{6}{1000}(385-55)=1.98$.
16	**(b)**	Intersect at $y=0,3$: $\int_0^3(3y-y^2)dy=\frac{9}{2}$.
17	**(a)**	$u=e^{-x}$: $-\int\tan u\,du = \ln\lvert\cos u\rvert + c$.
18	**(e)**	$u=x-2$: $\left[-2\sqrt{4-u^2}\right]_0^1 + \left[\arcsin\frac{u}{2}\right]_0^1 = 4-2\sqrt{3}+\frac{\pi}{6}$.